IOMETE Fit Analyzer: PRD

Working name: **FitLens**. Built for the IOMETE track at hackCBS (Make Complex Tech Easy to Buy). Version 0.1.

0. The honest verdict first

Your idea (company workflows in a .md file, AI says how relevant IOMETE is, dashboard plus export) is a **good fit for the track**, but in its raw form it is also the **most obvious idea** a team will have. Three problems to fix before you build:

1. **"ChatGPT with a nicer front end" risk.** The starter kit warns about this directly. If your app is one prompt ("rate IOMETE relevance for this company") plus charts, it will lose. The win is in structure: a curated, sourced capability base, a visible scoring rubric, and an evidence trail from every claim back to a line in the input and a documented IOMETE capability.
2. **A fit score is only useful if it can say no.** A tool that rates every company 8/10 is a hype machine. You must show one company that scores **low** and explain why. This is your strongest proof of accuracy (25% of the score).
3. **The dashboard is the least valuable part.** Execution is only 15%. Impact (35%), buyer clarity (25%) and accuracy (25%) are earned by the content of the report, not by chart count. Build the report first, the dashboard second, the export third.

Decide **one primary user**: a **sales rep qualifying an account and preparing a call**. The buyer-facing view ("how do I justify this internally") becomes an export variant of the same analysis, not a second product.

1. Problem

IOMETE is a sovereign data lakehouse: Spark and Iceberg on Kubernetes, deployable on premises, in the customer's own cloud, hybrid, or air gapped (from the starter kit). That is powerful but hard to map onto a prospect's actual business. A rep today must read the prospect's world, read IOMETE's docs, and mentally connect the two before a call. A buyer new to data platforms cannot tell whether "lakehouse" even applies to them.

The friction we remove: the gap between "here is what this company does" and "here is exactly where IOMETE is, and is not, relevant to it, with proof."

2. Users and jobs

User	Job to be done	What they need from the output
Sales rep (primary)	Qualify the account and prep the call in minutes	Fit verdict, top 3 use cases, likely objections, discovery questions, a talk track
Buyer, data or IT lead (secondary)	Understand if IOMETE applies and defend it internally	Plain-language explanation, what to read next, a short memo for their boss
Buyer, non-technical exec (secondary)	Understand value without jargon	One-page summary, no unexplained terms

3. Competitive landscape

I searched for current information. Treat everything marked *third party* as unverified marketing or aggregator content, not ground truth.

3.1 Who competes with the idea (sales and account research tools)

Category	Examples seen in current roundups	What they do well	Where they fall short for this problem
Account intelligence and intent	6sense, ZoomInfo, Demandbase	Predictive account scoring from fit and buying intent signals	Score *whether an account is in market*, not *whether this specific product maps to their workflows*. Opaque scores.
Enrichment and AI research workspaces	Clay, Apollo (agentic assistant launched March 2026), Octave, Aomni	Pull company data, run AI prompts per row, generate account briefs	Generic: no vendor-specific capability base, no "this workflow maps to this documented feature" evidence, no honest "unknown" handling by design

Category	Examples seen in current roundups	What they do well	Where they fall short for this problem
Conversation intelligence	Gong	Learn from recorded calls	Works after calls happen, not before
General LLMs	ChatGPT, Claude, Perplexity	Free, flexible, can read a pasted md file	No curated source of truth, inconsistent scores run to run, no structured export, confidently invents product claims

Honest read: the sales-tool market is crowded and well funded, and you will not out-feature it. Clay-style "AI fit scoring" prompts (score 1 to 10 on criteria you define) are already common practice. Your edge is not "AI scoring". It is **product-specific, source-grounded fit analysis**, which generic tools do not do because they have no deep model of one vendor's product.

3.2 Who competes for the buyer's attention (IOMETE's market, for use-case framing)

Third-party aggregators list **Snowflake, Databricks and Cloudera** as common IOMETE alternatives, and other lakehouse listings place **Dremio and Starburst** in the same category. Buyers in these deals compare on: governance depth, connector count, cost transparency, time to value, and operating burden. One third-party listing claims IOMETE has fewer pre-built connectors, thinner governance tooling and a smaller talent pool. That is an unverified third-party claim, but it tells you what objections a rep will hear. **Your tool should surface likely objections, not hide them.**

3.3 What others do that you should copy and beat

- Vendor maturity assessments and TCO calculators (typical at large data platform vendors; I did not verify specific ones): they work because they give a *score plus next step*. Beat them by making the rubric visible and editable instead of a lead-capture form.
- RFP answering tools: they win on "answer only from approved sources". Copy that discipline.

3.4 Your differentiation (say this to judges)

1. **Source-grounded:** every IOMETE claim links to a documented capability in a curated knowledge base. No link, no claim.

2. **Transparent math:** the score comes from a visible rubric, not an LLM vibe. Users can change weights and watch the score move.
3. **Honest unknowns:** missing information is a first-class output ("we could not tell X; ask this question"), not hidden.
4. **Can say no:** includes disqualifiers and "poor fit" verdicts with reasons.
5. **Two audiences, one analysis:** rep brief and buyer memo from the same evidence.

4. Product concept

4.1 End-to-end flow

1. **Input:** upload a .md file describing the company (workflows, systems, data sources, teams, constraints). Offer a downloadable **template** so the input is consistent. Optionally paste text instead of uploading.
2. **Choose audience:** Sales rep or Buyer. This changes tone and sections, not the underlying analysis.
3. **Parse:** extract structured workflows, data sources, tools in use, scale hints, compliance or residency constraints, pain points.
4. **Match:** map each workflow to IOMETE capabilities from the knowledge base.
5. **Score:** apply the fixed rubric to produce an overall fit and per-dimension scores.
6. **Report:** dashboard on screen plus an exportable report (PDF and Markdown).
7. **Next step:** concrete actions (questions to ask, docs to read, a suggested pilot).

4.2 Input design (fixing your biggest weakness)

Your plan depends on a user having a well-written md file about a company. Real reps will not write one. Improvements, in priority order:

- **P0:** provide a short md template with headings (Business, Workflows, Data sources, Current stack, Team, Constraints, Pain points) and a filled example.
- **P0:** an **input quality check**: before scoring, show "we found 6 workflows, 2 data sources, no mention of data volume or compliance". Low coverage means lower confidence and visible questions.
- **P1:** allow a bare company description or pasted text, not only a file.
- **P2:** auto-draft the md from a public company's website or filings (reuse the sourced/assumed labeling idea from the starter kit's call prep example).

5. Scoring model

The LLM extracts and classifies. **Code computes the score.** This is the main accuracy move.

5.1 Fit dimensions (draft weights, user-adjustable)

Dimension	What it asks	Why it matters for IOMETE	Weight
Sovereignty and deployment need	Must data stay in own infrastructure, on premises, or air gapped?	Core positioning: runs in the customer's own environment	25
Data scale and variety	Large, mixed structured and unstructured data?	Lakehouse value grows with volume and variety	15
Lock-in and openness concern	Worry about proprietary formats or vendor lock-in?	Built on open Spark and Iceberg	15
Analytics and AI workloads	BI, SQL, data engineering, ML pipelines?	Capability coverage	15
Cost pressure	Is current warehouse or cloud spend a pain?	Common buying trigger (verify against docs and pricing before claiming savings)	10
Technical readiness	Spark, Iceberg or Kubernetes familiarity, data team exists?	Determines time to value	10
Governance and compliance needs	Catalog, access control, audit requirements?	Fit and also a known objection area	10

5.2 Rules that make it trustworthy

- Each dimension score (0 to 100) is computed from tagged **evidence items**. Each evidence item has: source (md line or section), IOMETE capability id, direction (supports or hurts fit), strength (low, medium, high), and a confidence flag.
- **No evidence means "unknown"**, not a neutral score. Unknown dimensions are excluded from the average and shown as gaps, and they lower the **overall confidence** rating.
- **Disqualifier flags** (examples: no data team and tiny data volume; hard requirement for a managed SaaS only; no infrastructure control) cap the overall verdict regardless of other scores.

- Verdict bands: Strong fit, Possible fit (needs discovery), Weak fit, Not a fit. Always shown with a confidence level next to it.
- Run the same input twice and the score must match within a small tolerance. Use temperature zero for classification and cache extracted evidence.

6. Knowledge base (the core asset)

Build a hand-curated file, `iomete_capabilities.json`, of roughly 25 to 40 entries from IOMETE's own docs, University and blog. Each entry has:

- `id`, `name`, plain-language description (no jargon, written for a first-year student)
- `doc_url` (the exact page that supports it)
- `fits_when` and `does_not_fit_when` conditions
- `common_objections` and a safe, honest response
- `maturity_note`: if unsure about a capability, mark it **unverified** instead of guessing

Starting facts you can already cite from the starter kit: sovereign lakehouse, runs on premises, in the customer's cloud, hybrid or air gapped; built on Apache Spark and Apache Iceberg; runs on Kubernetes; open formats and no lock-in; built for enterprise data teams. Other capabilities (SQL editor, catalog, built-in BI, serverless Spark jobs, notebooks) appear in third-party listings; **confirm each against IOMETE's own docs before adding it** and put the doc link on the entry.

Why curated and not RAG over the whole docs: it is faster to build, easier to verify line by line, and prevents the model from inventing features. If time allows, add a retrieval layer later.

7. Report and dashboard spec

7.1 Report sections (this is the product)

1. **Verdict card:** fit band, overall score, confidence, one-sentence reason.
2. **Why this verdict (plain language):** 3 to 5 sentences a non-technical person can follow, with every IOMETE claim linked to its doc.
3. **Workflow map:** table of each company workflow, matched IOMETE capability, fit rating, evidence link, confidence.
4. **Top use cases:** ranked, each with the company's own words quoted from the input and the supporting doc link.
5. **Poor fit and risks:** disqualifiers, likely objections, and honest responses.
6. **Unknowns:** what we could not determine and the exact question to ask to resolve it.

7. **Competitive context:** where Snowflake, Databricks or others are likely incumbents, framed as questions to ask, not attack lines.
8. **Next steps:** 3 concrete actions (docs to read, discovery questions, suggested pilot scope).
9. **Audience extra:** for reps, a one-page talk track and call agenda; for buyers, a draft internal memo and the sections a CFO or security lead will ask about.
10. **Assumptions and method:** weights used, rubric version, what was inferred versus stated.

Every statement carries a label: **Stated** (from the input), **Sourced** (from IOMETE docs, with link), or **Assumed** (our inference). This is directly aligned with the judging criterion on accuracy.

7.2 Dashboard (P1, keep it simple)

- Overall fit gauge with confidence badge
- Radar chart across the seven dimensions
- Workflow heat table (workflow, capability, fit, confidence), clickable to see evidence
- Unknowns panel with the question per gap
- **What-if sliders** for dimension weights with the score updating live
- Export buttons

7.3 Export

- **Markdown:** easiest and always works.
- **PDF:** render the same markdown to HTML then PDF. Keep the layout clean and printable; a single-column report beats a pretty broken one.
- Include method and sources in the export so a finance or security reviewer can verify it.

8. Architecture

```
.md upload --> Parser (LLM, JSON schema) --> Input quality check
                                  |
                                  v
            Matcher (LLM, per workflow, against capability KB)
                                  |
                                  v
          Evidence items --> Scoring engine (pure code, versioned rubric)
                                  |
                                  v
   Report writer (LLM, only allowed to cite evidence ids) --> Validator
                                  |
                                  v
                Dashboard (UI) + Export (MD, PDF)
```

- **Stack suggestion for speed:** Python, with Streamlit and Plotly for the UI unless your team is much faster in React. Streamlit gets you upload, charts, sliders and download buttons in hours.
- **Models:** a smaller, cheaper Claude model for parsing and matching, a stronger one for the final narrative (confirm current model names and limits in the Anthropic docs).
- **Validator (important):** a code check that fails the report if any claim has no evidence id or no doc link, or if a score cannot be recomputed from the evidence items. Failing items are downgraded to Assumed or removed.
- **Reproducibility:** fixed prompts, temperature zero, cached evidence, rubric version stamped on every report.

9. Proving it works (what judges will look for)

Build three sample company md files before you build anything else, and **write down the expected verdict first**:

Sample	Purpose	Expected verdict
Regulated bank or defense-style org with on-premises or air gapped requirements and large mixed data	Should score high and explain why	Strong fit
Small startup on a single managed Postgres, no data team, tiny data	Negative control, must score low	Weak or not a fit

Sample	Purpose	Expected verdict
Mid-size media or retail company already on a cloud warehouse, mixed signals	Should land in the middle and produce good discovery questions	Possible fit

Report these numbers in your submission:

- **Verdict agreement:** your three expected verdicts versus the tool's output.
- **Citation coverage:** percent of IOMETE claims with a doc link (target 100 percent).
- **Unknown handling:** number of unknowns surfaced and the question for each.
- **Repeatability:** same input run 3 times, score range.
- **Time saved:** minutes a rep spends doing this manually versus the tool (measure yourself on one case; do not invent market-wide figures).

Do not claim savings or ROI numbers for IOMETE unless a documented source supports them.

10. Improvements to your original idea

Weakness in the original plan	Improvement
Garbage in, garbage out: the md file quality decides everything	Template plus input quality check plus confidence score that drops when information is missing
One opaque "relevance" number	Seven dimensions, visible weights, what-if sliders, evidence per score
Likely to flatter IOMETE	Disqualifiers, poor-fit verdicts and a negative-control sample
Risk of invented product claims	Curated capability KB with doc links and an "unverified" state
Report is a data dump	Audience-specific sections, plain language, concrete next step
Not clearly tied to a sales workflow	Output includes discovery questions, objections, talk track
Dashboard-heavy scope	Report first, dashboard second, PDF export third

Weakness in the original plan	Improvement
Only helps reps	Same evidence powers a buyer memo (the "defend this to my boss" friction)

11. Scope and build order

P0 (must ship)

- Capability KB of 20+ sourced entries
- md template, upload, parser, input quality check
- Matcher, evidence items, scoring engine with visible rubric
- Report with Stated, Sourced, Assumed labels, unknowns and next steps
- Markdown export
- Three sample companies with expected verdicts

P1 (should ship)

- Dashboard with gauge, radar, workflow table, what-if sliders
- PDF export
- Audience toggle (rep talk track versus buyer memo)
- Validator that blocks unsupported claims

P2 (stretch, cut first)

- Pasted-text or company-name input
- Competitor-incumbent detection
- Shareable link
- Retrieval over the full docs

Suggested time order if you are short on time: KB and samples, then parser and matcher, then scoring and report, then Markdown export, then dashboard, then PDF, then polish. If you must cut, cut from the bottom of that list, never from the KB or the validator.

12. Risks

Risk	Mitigation
Model invents IOMETE features	Closed capability list, validator, unverified state
Scores differ run to run	Temperature zero, cached evidence, code-computed score

Risk	Mitigation
Everything scores high	Negative-control sample, disqualifier caps, show it in the demo
Judges see it as a generic LLM wrapper	Lead the demo with the evidence trail and the low-score case
Weak or short input	Quality check, confidence drop, explicit questions
Time runs out	Markdown export and report before dashboard and PDF
IOMETE facts become outdated	Doc links on every entry, rubric and KB version stamped on exports

13. Two-minute demo script

1. State the friction in one line: complex tech is hard to place against a real company.
2. Upload the bank sample. Show the verdict, the workflow table, and click one claim to its doc link.
3. Move one weight slider and show the score change. Say that the math is visible.
4. Upload the startup sample. Show a low fit and the honest reason. Say the tool can say no.
5. Show the unknowns panel and the exact question a rep should ask.
6. Export the PDF and show the Assumptions and Method page.
7. Close with the numbers: verdict agreement, citation coverage, repeatability.

14. Submission checklist against the rubric

Criterion	Weight	What to show
Practical impact	35%	Rep goes from account description to discovery questions and talk track in minutes; before and after on one real example
Buyer clarity	25%	Plain-language verdict, no unexplained terms, clear next step
Accuracy	25%	Stated, Sourced and Assumed labels, doc links, unknowns, negative-control result
Working execution	15%	The full flow runs end to end on all three samples

15. What I could not verify

- IOMETE's exact current feature set, pricing and limits: confirm against docs.iomete.com before adding any capability to the KB.
- Third-party claims about IOMETE's connectors, governance depth and market position come from aggregator sites and may be outdated or AI-generated.
- Specific features and pricing of the sales tools listed above change often; the comparison is category-level, not a feature audit.
- Whether your hackathon allows submission to more than one track.